

◇ WEATHER INTELLIGENCE PLATFORM

WlaaS

Weather Intelligence as a Service

Transforming raw forecasts into autonomous, physical mitigation strategies.

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◇ DIAGNOSTICS

The Operational Disconnect

FIG. 02

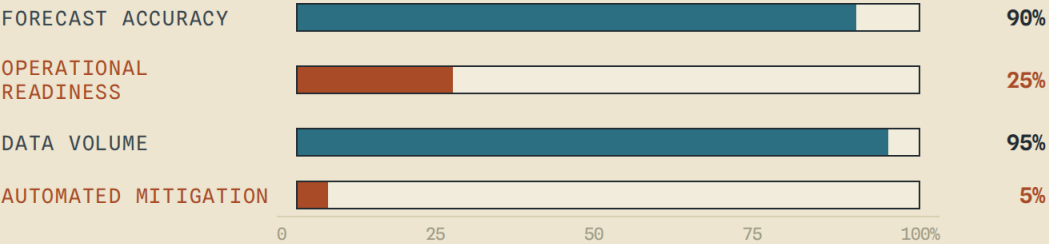
01

THE INDUSTRIAL REALITY

- ◇ **Observation vs. action.** Current platforms predict numbers — "it will be 42°C" — and leave humans to guess the operational impact.
- ◇ **Systemic vulnerability.** Grid failures, agricultural losses, and logistics disruptions occur rapidly, often faster than manual response cycles.
- ◇ **The core issue.** A massive disconnect exists between atmospheric physics and immediate operational decision-making.

02

CURRENT INDUSTRY CAPABILITY GAP



◇ RESPONSE ARCHITECTURE

The WlaaS Solution

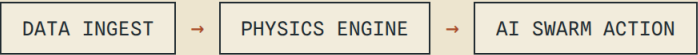
FIG. 03

01

THE BRIDGE

An end-to-end platform that fuses a deterministic physics engine (**FastAPI**) with a collaborative AI swarm (**n8n**) — converting raw atmospheric signal into coordinated field action.

DETERMINISTIC PIPELINE



02

ACTION OVER OBSERVATION

WlaaS does not just tell you what will happen — it applies environmental equations and autonomously orders immediate mitigation actions.

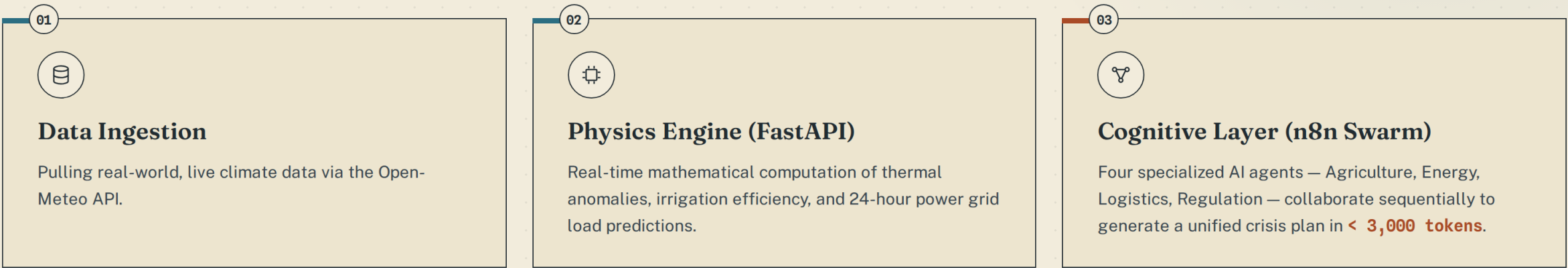
EXAMPLE — TRIGGERING MITIGATION VIA VAPOR PRESSURE DEFICIT

$$VPD = e_s - e_a$$

◇ STACK OVERVIEW

System Architecture

FIG. 04



◇ TOKEN OPTIMIZATION

Token-Efficient Architecture

FIG. 05

01

OPERATING PRINCIPLE

Rules, APIs and memory handle predictable work first. The LLM is activated only when interpretation, correlation or action generation is genuinely required.

02

DETERMINISTIC CORE

Intent Router

Classifies greeting, FAQ, direct metric or full analysis before any model call.

Zero-Token Regex Router

Uses keyword and pattern rules for predictable requests without an LLM.

Conditional Agent Activation

Runs only the specialist required by the selected route.

Fast Path vs Deep Path

Direct answers stay lightweight; full multi-agent analysis runs only when requested.

03

CONTEXT COMPRESSION

Compact State Vector

Reduces large backend payloads to only the metrics needed for reasoning.

API Normalization

Converts mixed API outputs into one small, consistent evidence structure.

Relevance Filtering

Removes distant or unrelated events before evidence reaches the model.

Limited Agent Context

Each agent receives only task-relevant fields instead of the full history.

04

MODEL & OUTPUT CONTROL

Fireworks AI + MiniMax-M3

Used only for specialist reasoning, evidence correlation and action generation. Fast inference and reliable instruction-following support concise, structured answers without routing every request through an LLM.

Session Memory + Cached Routes

Stores the region, previous assessment, stakeholder and evidence. Follow-up questions reuse prior results instead of repeating API calls or regenerating the same analysis.

Strict JSON + No-Invention Rule

Fixed fields control verbosity and make frontend parsing predictable. When evidence is insufficient, the workflow returns a safe fallback instead of spending tokens on speculation.

WHY THIS MODEL: a practical balance of speed, reasoning quality and token economy for real-time n8n orchestration.

05

TOKEN-EFFICIENT FLOW

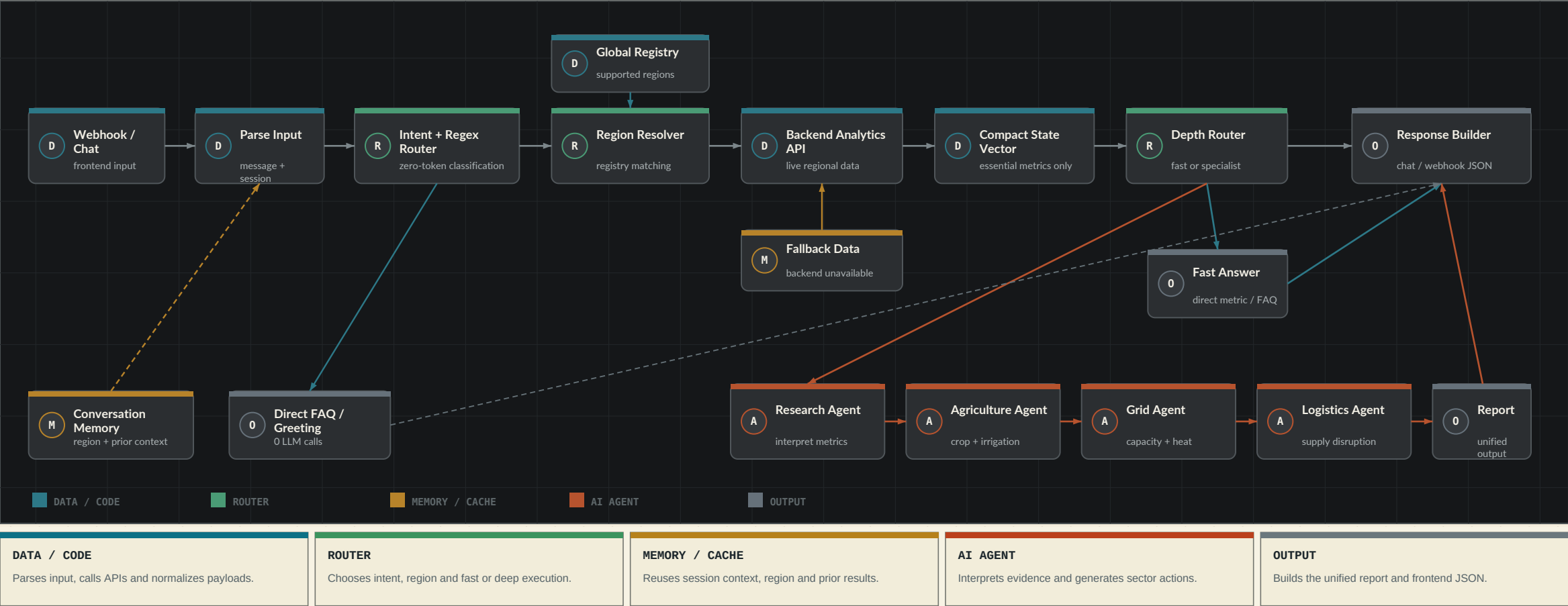
USER REQUEST -> DETERMINISTIC ROUTE -> COMPACT EVIDENCE -> FIREWORKS AI / MINIMAX-M3 ONLY IF NEEDED -> STRICT JSON RESPONSE

WORKFLOW MAP

WlaaS Assistant Workflow

Simplified logical view of the production architecture. The full WlaaS Assistant contains 60 n8n nodes; repeated validation, parsing, fallback, memory, API, agent and response steps are grouped into representative blocks below.

60 PRODUCTION NODES

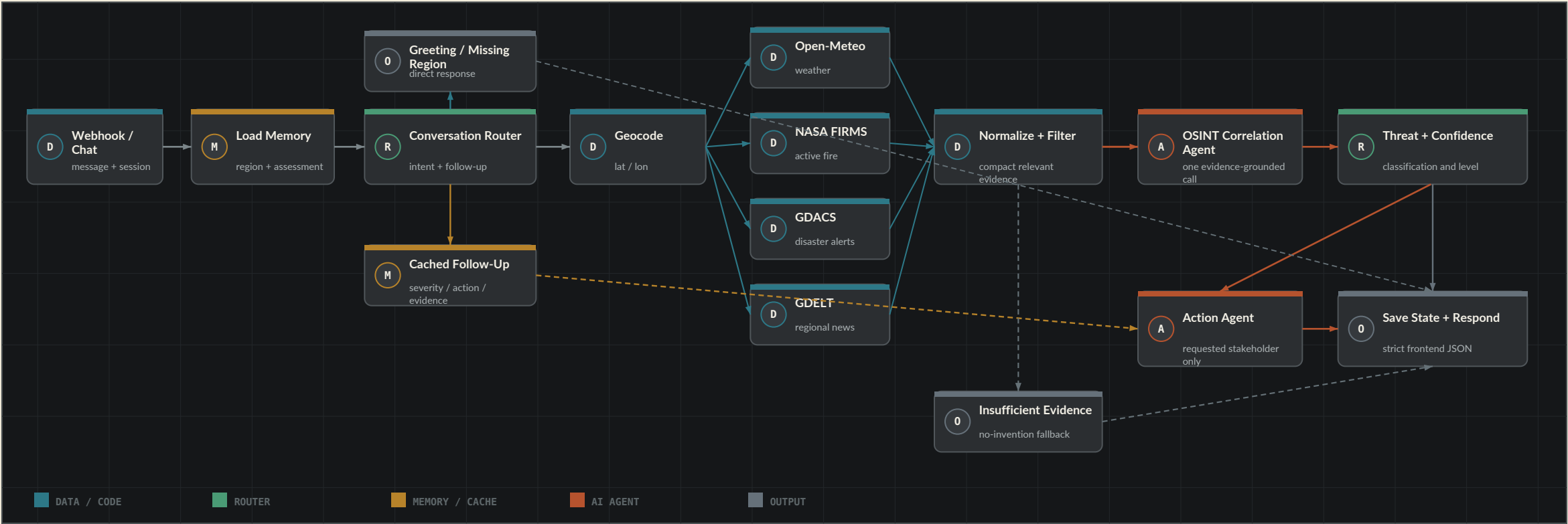


WORKFLOW MAP

CrisisLens Workflow

Simplified logical view of the production architecture. The full CrisisLens workflow contains 50 n8n nodes; repeated session, geocoding, parallel evidence, filtering, confidence, action and response steps are grouped below.

50 PRODUCTION NODES



INPUT / MEMORY	ROUTING / GEO	EVIDENCE	AI REASONING	OUTPUT / SAFETY
Loads message, session and the previous assessment.	Resolves intent, region and coordinates.	Runs parallel sources, then filters relevant events.	Correlates evidence, confidence and stakeholder actions.	Saves state, returns strict JSON or safe fallback.

◇ LIVE SIMULATION

Jaipur Heatwave

FIG. 05

01

ACTIVE CRISIS ANALYSIS

Deterministic prediction of a **15% grid blackout risk** at 7:00 PM, driven by compounding thermal anomalies.

AUTONOMOUS EXECUTION

Automatic triggering of protocols for immediate human work restrictions and preventative fuel load staging.

GLOBAL SCALABILITY

Integrated region registry capable of mapping, resolving, and simulating risk across **420+ cities** instantly.

BLACKOUT RISK WINDOW

19:00 LT

SIMULATED REGION COVERAGE

420+ Cities

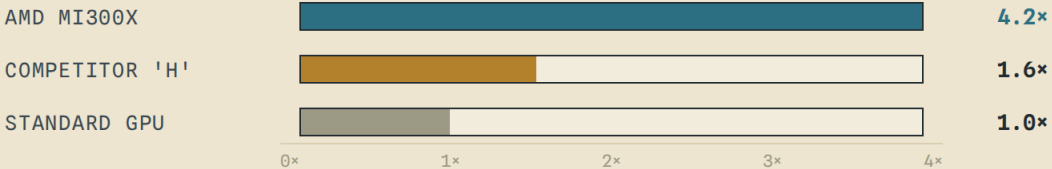
◇ HARDWARE PARTNER

Powered by AMD Instinct™ MI300X

FIG. 06

01

RELATIVE SIMULATION THROUGHPUT



Throughput advantage driven primarily by the 192GB HBM3 VRAM capacity, enabling zero-latency co-location of physics and cognitive models.

02

BREAKING THE COMPUTE BOTTLENECK

Traditional processors saturate under real-time geospatial physics and multi-agent reasoning. The AMD CDNA™ 3 architecture provides massively parallel processing.

03

CO-LOCATING PHYSICS & COGNITION

Running local swarms alongside deep-physics models demands immense memory. The MI300X delivers **192 GB** of HBM3 VRAM and **5.3 TB/s** bandwidth.

04

ZERO-LATENCY SYNERGY

The entire GNN pipeline and agentic models load onto the exact same card, eliminating PCIe bottlenecks.

◇ ENGINEERING DISCIPLINE

The WlaaS Advantage

01

ABSOLUTE RELIABILITY

Zero AI hallucinations. Every agent response is mathematically constrained and strictly guided by real metrics from the backend physics engine.

02

OPTIMIZATION AT SCALE

A complex, multi-agent operational architecture is compressed into less than **3,000 tokens** per full run, achieving extreme efficiency.

03

TOKEN USAGE – PER FULL SIMULATION RUN

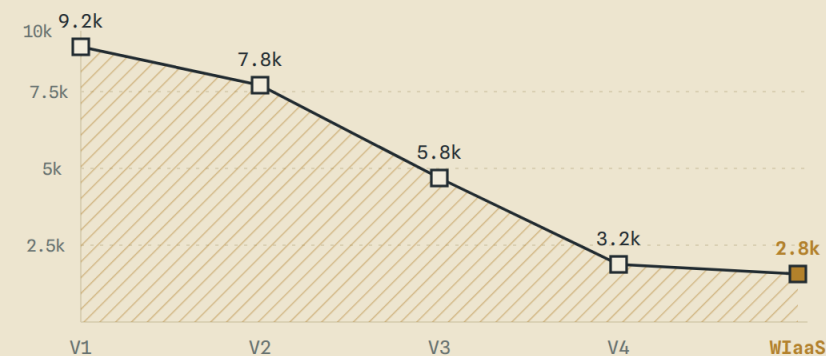


FIG. 07

◇ COMMERCIAL MODEL

Business Model & Economy

FIG. 08

01

MONETIZATION STRATEGY

B2B SaaS & API. Tiered subscription model for enterprise and grid operators, alongside a pay-per-simulation API for AgTech platforms.

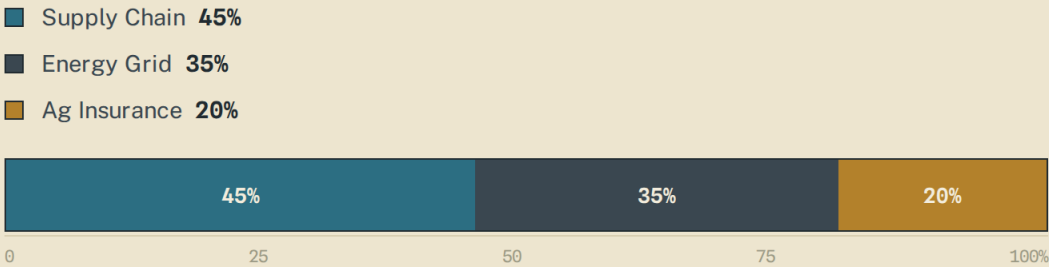
"Mitigating physical climate risk is no longer a luxury; it is a baseline requirement for scaling businesses."

TARGET SECTORS

- ◇ **Supply Chain** — protecting high-value perishables.
- ◇ **Energy Grid** — proactive load-shedding intelligence.
- ◇ **Ag Insurance** — parametric micro-insurance payouts.

02

REVENUE COMPOSITION



◇ FORWARD TRAJECTORY

Roadmap & Vision

FIG. 09

